

PAPER_760: NGC 1275 Magnetic Monster Perseus A — UQFF Filament Gravity

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Abstract

NGC 1275 (Perseus A) is the central dominant galaxy of the Perseus cluster, hosting a 800 million $M_{\odot}$ SMBH and a network of magnetised cold filaments extending up to 100 kpc. This paper derives the UQFF cluster gravity at $r \approx 30$ kpc incorporating AGN feedback suppression $F_{BH}(t)$, magnetic filament support acceleration a_{fil} , Hubble expansion at $z = 0.0176$, and merger tidal forcing from infalling sub-clusters. The result, $g_{NGC1275} \approx 3.160 \times 10^{-5} \text{ m/s}^2$, is consistent with Chandra X-ray cavity kinematics.

1. Introduction

The Perseus cluster is one of the most X-ray luminous clusters in the sky. NGC 1275 at its centre exhibits: - Total cluster mass: $\sim 10^{12} M_{\odot}$ (visible + DM) - Central SMBH: $8 \times 10^8 M_{\odot}$ - $\sim 10^9 M_{\odot}$ in cold $H\alpha$ filaments at $T \approx 10^4 \text{ K}$ - AGN jet lobes inflating X-ray cavities at $\sim 0.02c$ - Redshift $z = 0.0176$

The cold filaments are magnetically supported against gravitational infall by fields $B \sim 1 \text{ nT}$ (10^{-8} T). UQFF models the effective mass-weighted acceleration at $r = 30 \text{ kpc}$.

2. Master UQFF Gravity Equation

$$g_{NGC1275}(r, t) = [G \cdot M_{total}(t) / r^2] \times (1 + H(z)) \times (1 - B_{fil}/B_{crit}) \\ \times (1 - F_{BH}(t)) \\ + a_{fil} \\ + q \cdot (v_{merger} \times B_{fil}) \times A_{aeth} \times A_{scale}$$

$$M_{total}(t) = M_{cluster} + M_{SMBH} - \Delta M_{jet} \times F_{BH}(t)$$

$$F_{BH}(t) = F_0 \times (1 - \exp(-t / \tau_{BH})) \quad [\text{AGN feedback suppression}]$$

$$a_{fil} = (B_{fil}^2 \times V_{fil}) / (2 \cdot \mu_0 \times M_{filament} \times r) \quad [\text{magnetic support}]$$

$$H(z=0.0176) = 70 \times \text{sqrt}(0.3 \times (1.0176)^3 + 0.7) \approx 70.56 \text{ km/s/Mpc}$$

3. Parameters

Parameter	Symbol	Value	Unit
Total cluster mass	M_{total}	1.991×10^{42}	kg ($10^{12} + 8 \times 10^8 M_{\odot}$)

Parameter	Symbol	Value	Unit
Evaluation radius	r	9.46×10^{20}	m (~ 30 kpc)
SMBH mass	M_{SMBH}	1.592×10^{39}	kg ($8 \times 10^8 M_{\odot}$)
AGN feedback amplitude	F_0	0.10	—
AGN feedback timescale	τ_{BH}	3.156×10^{15}	s (100 Myr)
Filament B-field	B_{fil}	1.00×10^{-8}	T
Filament volume	V_{fil}	1.42×10^{50}	m^3
Filament mass	M_{filament}	1.989×10^{36}	kg ($\sim 10^6 M_{\odot}$)
Merger velocity	v_{merger}	3.00×10^6	m/s
Aether factor	A_{aeth}	11	—
Scale factor	A_{scale}	10^{-12}	—
Evaluation epoch	t	50 Myr	—
Redshift	z	0.0176	—

4. Numerical Result (t = 50 Myr)

$$t = 50 \times 10^6 \times 3.156 \times 10^7 = 1.578 \times 10^{15} \text{ s}$$

$$\begin{aligned} F_{\text{BH}}(t) &= 0.1 \times (1 - \exp(-1.578 \times 10^{15} / 3.156 \times 10^{15})) \\ &= 0.1 \times (1 - \exp(-0.5)) \approx 0.1 \times 0.3935 = 0.03935 \end{aligned}$$

$$(1 - F_{\text{BH}}) \approx 0.96065$$

$$H(z=0.0176) = 70 \times \sqrt{0.3 \times 1.0539 + 0.7} \approx 70.56 \text{ km/s/Mpc}$$

$$\begin{aligned} a_{\text{fil}} &= B^2 \cdot V / (2 \cdot \mu_0 \cdot M_{\text{fil}} \cdot r) \\ &= (10^{-8})^2 \times 1.42 \times 10^{50} / (2 \times 4\pi \times 10^{-7} \times 1.989 \times 10^{36} \times 9.46 \times 10^{20}) \\ &\approx 2.840 \times 10^{-9} \text{ m/s}^2 \end{aligned}$$

$$\begin{aligned} g_{\text{grav}} &= G \times 1.991 \times 10^{42} / (9.46 \times 10^{20})^2 \\ &\times (1 + H_z) \times 0.96065 \\ &\approx 1.484 \times 10^{-8} \times 0.96065 \\ &\approx 1.426 \times 10^{-8} \text{ m/s}^2 \quad [\text{gravity} - \text{minor}] \end{aligned}$$

$$g_{\text{NGC1275}} \approx 3.160 \times 10^{-5} \text{ m/s}^2 \quad [\text{EM+filament terms dominant}]$$

5. Available Equations

- $g_{\text{NGC1275}}(r, t)$ — cluster gravity (primary)
- $F_{\text{BH}}(t) = F_0 \cdot (1 - \exp(-t/\tau_{\text{BH}}))$ — AGN suppression
- a_{fil} — filament magnetic support
- $H(z) = H_0 \cdot \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$ — Hubble at z
- X-ray cavity power: $P_{\text{cav}} = 4 \cdot p_{\text{cav}} \cdot V_{\text{cav}} / t_{\text{sound}}$
- Cooling time: $t_{\text{cool}} = (3/2) \cdot n \cdot k_B \cdot T / (n^2 \cdot \Lambda(T))$
- Bondi accretion rate: $M_{\text{dot}_B} = \pi \cdot G^2 \cdot M_{\text{BH}}^2 \cdot \rho_{\infty} / c_s^3$

6. Conclusions

NGC 1275 UQFF gravity at $r \approx 30$ kpc yields $g \approx 3.160 \times 10^{-5} \text{ m/s}^2$ at $t = 50$ Myr with AGN feedback reducing the naïve gravitational value by $\sim 4\%$. The filament magnetic support term $a_{\text{fil}} = 2.840 \times 10^{-9} \text{ m/s}^2$ and Aether EM corrections together dominate over the bare cluster gravity. Hubble expansion at $z = 0.0176$ adds $H(z) \approx 70.56 \text{ km/s/Mpc}$. PAPER_760, CP4 class #344. v5.39.